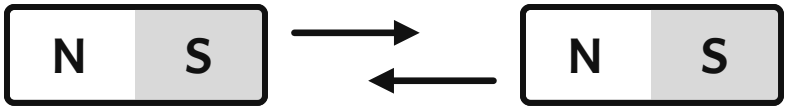
PHYSICAL AND CHEMICAL SCIENCE

Magnets and Electromagnets

Magnetic materials, poles, attraction and repulsion, magnet strength, compasses, electromagnets and uses



S and N: attract (pull together)



N and N: repel (push apart)

In this pack

- **Learn it:** Magnets and electromagnets ★☆☆
- **Fact check:** Attract or not? ★☆☆
- **Think it through:** Magnet puzzles ★★☆
- **Investigate:** How strong are our magnets? ★★☆
- **Real-world case study:** Giant magnets in the ship-breaking yards ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C7 · MAGNETS

Magnets and electromagnets

Name: Class: Date:

Learning goals

- Observe how magnets behave and which materials they attract.
- Describe magnetic poles and how magnets attract and repel.
- Explain how electromagnets work and how magnets are used every day.

More than two thousand years ago, people found strange stones that could pull bits of iron towards them. They called them **lodestones**. Hung on a thread, a lodestone always turned to point north. These were the first **magnets**, and the first **compasses** helped sailors cross the oceans.

What do magnets attract?

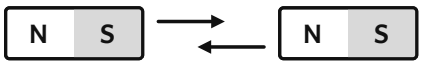
Magnets **attract** (pull) things made of **iron**, **steel** (which contains iron), **nickel** and **cobalt**. These are **magnetic materials**. Most other metals are **not** magnetic: aluminium, copper, brass, gold and silver. Non-metals such as wood, plastic, glass and paper are not magnetic either.

Remember

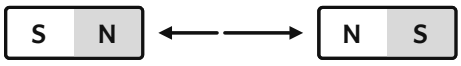
Not all metals are magnetic! A magnet picks up a steel paper clip, but not an aluminium can or a copper wire.

Poles

Every magnet has two ends called **poles**: a **north (N) pole** and a **south (S) pole**. The magnetic force is strongest at the poles.



S and N: attract (pull together)



N and N: repel (push apart)

Like poles repel; unlike poles attract.

- **Unlike** poles (N and S) **attract** each other.
- **Like** poles (N and N, or S and S) **repel** (push each other away).

Magnetic force works at a distance

Magnetism is a **non-contact force**. A magnet can pull a paper clip without touching it, and the force passes through paper, plastic, water and even your hand. The space around a magnet where it has an effect is its **magnetic field**.

Strong and weak magnets

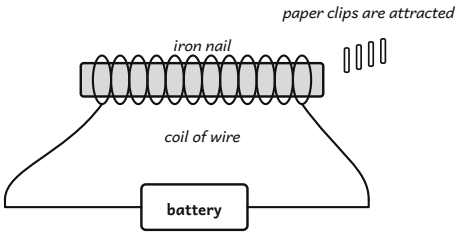
Some magnets are stronger than others. A stronger magnet can hold more paper clips, or pull a paper clip through more sheets of paper. Size does not always tell you the strength: some tiny magnets are very strong.

Earth is a giant magnet

The Earth behaves like a huge magnet. A **compass** needle is a small magnet that swings round to point **north**. Fishers on the Bay of Bengal, pilots and hikers use compasses to find their way.

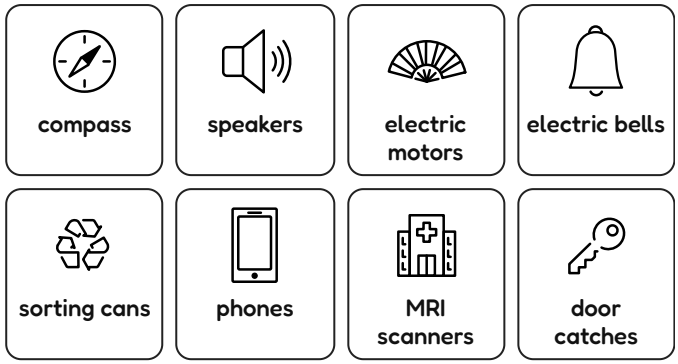
Electromagnets

Wrap a long wire many times around an **iron nail** and connect it to a battery. Electricity flowing through the coil turns the nail into a magnet: an **electromagnet**. Switch off the electricity and the magnetism disappears. More coils or more cells make a stronger electromagnet.



A simple electromagnet.

Magnets in everyday life



Fridge doors use magnetic strips to stay closed. Recycling plants use magnets to pull steel cans out of rubbish. Huge electromagnets on cranes lift scrap iron and drop it when the current is switched off.

⚠ Stay safe

Keep strong magnets away from phones, bank cards and computers, and never let small children swallow magnets.

Key words

magnet	an object that attracts iron and steel
magnetic material	a material attracted to a magnet: iron, steel, nickel, cobalt
pole	one end of a magnet, north or south
attract	pull towards
repel	push away
magnetic field	the space around a magnet where it has an effect
compass	a magnetic needle that points north
electromagnet	a magnet made by electricity flowing in a coil

FACT CHECK · C7 · MAGNETS

Attract or not?

★☆☆
Grade 4

Name: Class: Date:

Learning goals

- Predict which materials are magnetic.
- Use pole rules.

1 Sort the objects. [8]

SORT THESE

plastic ruler **aluminium can** **iron nail** **steel paper clip** **wooden pencil**
steel spoon **gold ring** **copper wire**

Attracted to a magnet	Not attracted

2 Attract or repel? [4]

Poles facing each other	Attract or repel?
N and S	
N and N	
S and S	
S and N	

3 **True or false?** [4]

Statement	True	False	If false, correct it
1. All metals are magnetic.			
2. A magnet can attract a paper clip through paper.			
3. An electromagnet stops working when the electricity is switched off.			
4. A compass needle points north.			

4 **Fill in the gaps.** [4]

WORD BANK

plastic **compass** **attract** **wire** **repel** **poles**

The ends of a magnet are called _____. Like poles _____. A _____ helps us find north. An electromagnet is made with a coil of _____ around an iron nail.

Magnet puzzles

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Explain magnet behaviour.
- Use electromagnets to solve problems.

1 **Rafi has a bag of mixed aluminium and steel cans for recycling. How could he separate them quickly? Explain.** [2]

2 **A magnet has lost its N and S labels. How could you find which end is north, using a labelled magnet?** [2]

3 **Why does a scrap-yard crane use an electromagnet and not an ordinary magnet?** [2]

4 **Suggest two ways to make an electromagnet stronger.** [2]

5 **Draw a magnet picking up a chain of paper clips. Label the poles and show where the force is strongest.** [2]

Magnet and clips

INVESTIGATE · C7 · MAGNETS

How strong are our magnets?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Plan a fair test.
- Record and present results.
- Draw conclusions.

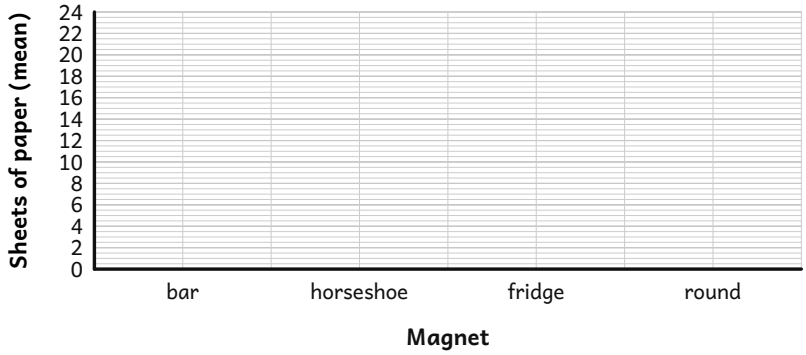
Class 4B tested four magnets. They counted how many sheets of paper each magnet could hold a steel paper clip through.

1 **Complete the plan.** [3]

Change	
Measure	
Keep the same	

2 **Complete the table and draw a bar chart.** [4]

Magnet	Try 1	Try 2	Try 3	Mean
bar magnet	12	14	13	
horseshoe magnet	18	17	19	
fridge magnet	3	3	3	
small round magnet	20	22	21	



3 **Which magnet was strongest? Was it the biggest?** [2]

.....

.....

4 **Why did they do three tries for each magnet?** [1]

REAL-WORLD CASE STUDY · C7 · MAGNETS

Giant magnets in the ship-breaking yards

★★★ Grade 4

Name: Class: Date:

Learning goals

- See how magnets are used in industry.
- Use numbers.
- Think about workers' safety and recycling.

Science | Economics | Environment | Ethics

Along the beach at **Sitakunda**, near Chattogram, old ships from all over the world are taken apart in some of the biggest **ship-breaking yards** in the world. The steel from the ships is cut into pieces and **recycled**. Much of the steel used for building in Bangladesh comes from these old ships.

In steel yards and scrap yards, cranes with huge **electromagnets** lift piles of iron and steel. When the operator switches the current on, the scrap jumps up to the magnet; switching it off drops the scrap onto a truck. Ship breaking gives many people jobs, but it can be **dangerous** for workers and can **pollute** the beach and sea with oil and chemicals, so there are rules to make it safer and cleaner.

Based on reports about the ship-recycling industry in Bangladesh.

1 **Why is an electromagnet useful for lifting scrap steel?** [2]

2 **Would the electromagnet pick up the copper cables and aluminium parts from a ship? Why?** [2]

3 **Maths. A crane magnet lifts 2 tonnes of steel each time. How many lifts are needed to load 30 tonnes?** [1]

4 **Recycling ship steel is good, but there are problems. Complete the table.** [2]

Good points	Problems

5 **Suggest two rules that would make ship breaking safer for workers.** [2]

.....

.....

Show what you know

★★★ Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

L	A	O	U	O	F	U	F	Y	K
C	B	T	A	M	U	K	Y	Y	F
O	U	K	T	S	A	C	A	W	R
M	A	I	D	R	T	G	B	I	O
P	P	O	L	E	A	E	N	G	D
A	I	R	P	D	C	C	E	E	I
S	S	O	U	T	H	D	T	L	T
S	I	H	M	R	E	P	E	L	P
D	K	W	O	L	I	R	O	N	W
F	E	N	O	R	T	H	P	G	N

attract compass iron magnet north pole repel south steel

2 Which object would a magnet attract? [1]

- ◯ a copper coin
- ◯ an aluminium can
- ◯ an iron nail
- ◯ a glass marble

3 Two north poles are pushed together. They will... [1]

- ◯ attract
- ◯ repel
- ◯ stick together
- ◯ become weaker

4 Design a game that uses magnets (e.g. a fishing game). Draw it and explain how the magnets work. [3]

My magnet game

5 **Challenge. Explain how you could make an electromagnet that is strong enough to lift 10 paper clips, and how you would test it fairly.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
predict which materials are magnetic	○	○	○
explain attract and repel	○	○	○
test the strength of magnets	○	○	○
explain how a compass works	○	○	○
make and explain an electromagnet	○	○	○
describe how magnets are used in industry	○	○	○